

DAVID XCHEL MORALES HURTADO

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Software Engineer with 5+ years of professional experience building automation, validation and infrastructure tooling on **Linux** platforms. Strong background in **Python**, **C++**, automation and large scale test frameworks. Experienced in **developing** reliable engineering solutions, **mentoring** and **collaborating** across **global organizations**. Passionate about **Open-Source** software, **Mathematics** and continuous learning. Always aiming towards high **Quality** and **performance** with a **mentoring mindset** in favor of the continuous growth of the team.

SELECTED ACHIEVEMENTS

- **Bronze Medalist** – Mexican Mathematics Olympiad.
- **Reduced validation execution times** from hours to minutes through **automation** using **Python** and **Bash**.
- **Improved testing efficiency** by 50% across processor validation programs.
- **Technical mentor** and **onboarding lead** for new engineers.
- **Contributed** on Open Source with: Fixing a [custom property type changing bug](#) on [Blender](#), an [overflow bug](#) fix on the **FMT** library, a [duplicate buffer deletion](#) on the **Godot Engine**, a [large data graph cycle check test](#) on the **CXXGraph** project, and on the **UnifyAI** python project test base.
- **International academic exchange** at Universiti Malaya.

PERSONAL PROJECTS

- Personal Arch Linux system, learning **Linux** and its **File System** deeply
- Personal **portfolio** found at dxchel.github.io, made using **React** and **TS**.
- Continuously evolving **C++** modular **GUI application** using **Gtk4** with **GTEST** testing. It currently supports an embedded internet browser, a fractal viewer using **Threading** and a LAN chat using **Coroutines**.

Senior Software Reliability Developer

February 2026 — Present

Oracle, Guadalajara

Develop **automation** and **cloud**-management tooling in **Python**, Java and Bash for Oracle Cloud Infrastructure regions.

- Develop and maintain **automation** workflows aimed at reducing manual effort, improving **reliability** and **accelerating** cloud-region deployment activities.
- Troubleshooting of **Linux** systems and **cloud** infrastructure issues.
- **Developed** AI-assisted tooling for information extraction, documentation and operational support.
- **Create** technical workflows and onboarding material used across engineering teams.

Functional Validation Engineer

February 2021 — February 2026

Intel, Guadalajara

Developed **Python** and **Bash** scripts made for the automated validation of next-generation processor architectures on **Linux** systems in collaboration with a **multicultural** and **diverse team** from all around the world.

- Helped **design** and **maintain** large-scale automation frameworks using **Python**, **Bash** and **PyTest**.
- **Reduced execution times from hours to minutes** through **automation** of manual workflows.
- **Created** migration tooling that reduced project-porting effort **from one day to minutes**.
- **Developed CI/CD**-based validation pipelines supporting multiple processor programs.
- Acted as **technical lead** and **point of contact** across teams.
- **Mentored** new engineers on processor architecture, memory systems and **Linux**-based development workflows.
- Improved validation **efficiency by 50%** through **optimization of test coverage** and execution strategies.

EDUCATION

Mechatronics Engineer

August 2015 — January 2021

Instituto Politecnico Nacional, Mexico City

I was always a **top performing student** in this prestigious institution with **effort**, **curiosity** and **creativity**. Always focused on **teamwork** for the best outcome by **mediating** and **leading** delivering **quality** projects **efficiently**.

Relevant projects include:

- **Python** projects for **Neural Networks**, **file** and **database processing**, **web scraping**, Flask applications with Selenium testing.
- **MIPS** implementation in **System-Verilog** with **UVM testbench** creation using **python** and **C++**.
- **Virtual reality** experiences using **Unity** and **blender**.
- Embedded systems and FPGA development projects.
- Electronic circuits design and implementation.

Mobility program

Universiti Malaya, Kuala Lumpur

Learned from one of the most renowned universities in the world while acquiring and adapting my experience working, **communicating** and **leading** people with contrasting backgrounds and mindsets.

Relevant projects include:

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Line follower with **Raspberry PI**.

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Hand tracking **AI** project.

February 2019 — June 2019

SKILLS

Languages

Python, C++, Bash, C, Java, JavaScript, TypeScript, Verilog, Assembly

Platforms

Linux(Ubuntu, Debian, Arch), **Git**/**GitHub**, **Docker**

Engineering

Agile, **CI/CD**, **Automation**, **Testing Frameworks**, **Mathematics**, AI-assisted Development, Embedded Systems, Cloud

Professional

Leadership, **Mentoring**, **Technical Communication**, **Problem Solving**

Languages

English, Spanish, learning German

COURSES

Advanced Python, Universidad Marista	June 2024 — November 2024
Linux Essentials, Linux Professional Institute	January 2020 — April 2020
Advanced C++ Programming and Modern Practices	February 2026 — May 2026
Time Management Fundamentals	February 2026 — March 2026
Foundations of Agile Project Management	February 2026 — March 2026
Learning how to Learn, Oakland University	January 2025 — June 2025
Mobile App Development, Google Activate	November 2019 — February 2020
Intel x86 Processor & Platform Architecture, MindShare	January 2022 — June 2022

EXTRA-CURRICULAR ACTIVITIES

Mexican Mathematics Olympiad, OMM, Mexico City Bronze award.	March 2013 — November 2014
Mexican Informatics Olympiad, COMI, Mexico City Participation award.	January 2014 — May 2014
Scout, Mexican Scout Association, Morelia Participation in altruism and teamwork oriented activities	January 2014 — August 2015
Robot war staff, IEEE university branch, Mexico City Staff in planning and organization of international event.	July 2018 — July 2020
Emergency Response Helper, ERUM, Mexico Provided help during the 2017 Mexico city earthquake	September 2017 — October 2017